

ETHIOPIAN HEALTH PROFESSIONALS LICENSING EXAMINATION(EHPLE) PRACTICE TEST BOOKLET 2025

ENVIRONMENTAL HEALTH



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MINISTRY OF HEALTH-ETHIOPIA

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HEALTHIER CITIZENS FOR PROSPEROUS NATION!



Message from the State Minister, Ministry of Health – Ethiopia



Mrs. Frehiwot Abebe

Improving the quality of healthcare remains a critical global priority and is essential for achieving sustainable development and universal health coverage. One of the most effective ways to ensure high standards in healthcare delivery is through the consistent assessment of professional competencies. In line with this commitment, the Ministry of Health of Ethiopia launched the Ethiopian Health Professionals Licensing Examination (EHPLE) in 2019 for undergraduates in seven core health disciplines. Since its inception, the program has expanded to encompass 17 health professions, reflecting our continued effort to build a competent and responsive health workforce.

The EHPLE was established with the primary aim of identifying and certifying health professionals who possess the minimum competencies required to provide safe and effective care. This ensures that all practicing professionals are equipped to meet the expectations of the communities they serve. The exam is coordinated by the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (LEO), which consists of four specialized desks. The LEO plays a vital role in system strengthening, enabling the exam to be conducted with increased efficiency, organization, and quality assurance.

It is important to recognize that the EHPLE is not a conventional academic or recruitment examination. Rather, it is a rigorous, competency-based assessment designed to evaluate the practical skills and critical thinking abilities that are essential for professional practice. This *Practice Items Booklet* is part of our broader strategy to support examinees and educators in preparing for the licensing process. It offers a realistic overview of the exam structure, promotes a deeper understanding of its objectives, and fosters transparency and sustainability in implementation.

The development of this booklet has been made possible through the collaborative efforts of experts from higher education institutions, the Addis Ababa University Testing Center, and professional associations. Their insights and dedication have been instrumental in shaping this resource. I extend my heartfelt thanks to all contributors and to the Ministry of Health team for their tireless work and steadfast commitment to excellence in healthcare regulation.

We believe this booklet will serve as a valuable tool for candidates preparing for the examination and for educators guiding them toward success. Together, we are building a stronger, more competent health workforce for Ethiopia.

Introduction EHPLE

Welcome to the 2025 Edition of the Ethiopian Health Professionals Licensing Examination Practice Test Booklet.

This resource is designed to support candidates in their preparation for the national licensing examination, ensuring they are well-equipped with the knowledge and clinical reasoning skills required to deliver competent and ethical healthcare across Ethiopia.

In Ethiopia, the licensing examination serves as a pivotal tool in evaluating the competency of health professionals who have completed their academic training and are ready to transition into independent practice. The exam not only ensures public safety by verifying professional readiness but also upholds national and international standards in healthcare service delivery. As the healthcare system continues to expand and evolve, a strong, well-regulated professional workforce becomes even more essential.

This practice booklet contains 200 multiple-choice questions modeled after the structure, content, and format of the actual licensing examination. The questions span across core clinical domains essential for entry-level practice. This includes areas such as Patient care, leadership and management, ethics, and Health Promotion as outlined in the Information Booklet 2024.

The booklet is intended to:

- Familiarize candidates with the structure and cognitive level of the actual licensing examination.
- Offer opportunities for self-assessment and identification of strengths and gaps in knowledge.
- Encourage critical thinking and clinical decision-making in scenarios reflective of real-world healthcare practice in Ethiopia.
- Reduce exam-related anxiety by providing realistic practice experiences.

Administered under the auspices of the Ministry of Health and relevant regulatory bodies, the Ethiopian Health Professionals Licensing Examination is grounded in evidence-based frameworks and aligned with both national needs and global best practices. It has been structured to assess a range of competencies including medical knowledge, communication, professionalism, and safe clinical practice.

This practice test is not a predictive tool, and performance on it does not guarantee similar outcomes on the actual exam. However, it provides an invaluable opportunity for familiarization with question styles, pacing strategies, and content review. We strongly recommend that candidates simulate real testing conditions while using this resource, including timing their responses and reviewing rationale for both correct and incorrect answers.

The goal of licensing examinations is not only to regulate entry into the profession but also to inspire excellence and lifelong learning. By engaging with this practice test, you are taking a proactive step toward professional development, accountability, and a commitment to serving your patients with competence and compassion.

We wish you success on your licensing journey—and beyond.

Acknowledgements

This *Practice Test Items Booklet for the Ethiopian Health Professionals Licensing Examination* is the result of a collaborative effort among dedicated professionals committed to advancing the quality of health sciences education in Ethiopia. The Ministry of Health extends its sincere appreciation to all those who contributed their time, expertise, and insights to the development of this essential resource.

Special recognition is given to the item developers and reviewers from various higher education institutions—both public and private—whose academic rigor and practical experience were instrumental in shaping the content and ensuring its relevance to real-world healthcare practice. Their contributions have helped create a high-quality, standardized practice tool that reflects national competencies and global benchmarks.

We are especially grateful to the team at the Addis Ababa University Testing Center, whose technical support and academic guidance were critical throughout the process. Their leadership in assessment design and educational evaluation has significantly strengthened the quality of this booklet.

We also thank the committed staff of the Ministry of Health, particularly those within the Health and Health-Related Institutions and Professionals' Regulatory Lead Executive Office (HHRIPR LEOs), whose coordination, oversight, and unwavering dedication ensured the successful completion of this project.

Together, these collective efforts reflect our shared vision of building a competent, ethical, and resilient health workforce for Ethiopia.

Environmental Health

1. Due to an expected flood in the upcoming summer in the frequently flood-affected rural community, people were displaced to a nearby open area for a temporary stay. For fear of sanitation-related diseases, an environmental health expert from the zonal health department was assigned to find an emergency excreta management facility for the displacement area.

What is the most appropriate sanitation facility for this case?

- A. Pit latrine
- B. Bucket latrine
- C. Trench latrine
- D. Compost latrine

ANSWER: C

2. An expert from a nearby municipality was assigned to advice residents of a condominium building on a liquid waste management option. They were looking for a suitable containment that can digest the organic content and reduce the solids for the human excreta from the water closets and the gray water from sinks without odor problems when released into a seepage trench.

What is the most appropriate treatment technology for the above case?

- A. Cesspool
- B. Septic tank
- C. Trickling filter
- D. Stabilization pond

ANSWER: B

3. As a response for the complaint by the residents of a rural village, an environmental health expert of the district conducted a sanitation facility satisfaction and utilization survey. The result of the study showed that most complaints are from the traditional pit latrine users. Consequently, the expert decided to upgrade all the traditional pit latrines in the community to ventilated improved pit latrines.

What is the most likely problem which can be improved by the upgraded VIP latrines?

- A. Simplicity
- B. Odor problem
- C. Cost of construction
- D. Groundwater contamination

ANSWER: B

4. Concerned experts were discussing defining improved sanitation facilities in a rural village context. Finally, they agreed that a latrine with a washable slab can be considered as an improved sanitation facility.

What is the most likely explanation for the experts' decision?

- A. It offers privacy for the user
- B. It ensures the user can safely use
- C. It separates the waste and the user
- D. It minimized the cost of construction

ANSWER: C

5. A sanitation promotion officer, while promoting ecological sanitation to rural community, emphasized the strict separation of urine from the feces for functionality of the system.

What is the most likely explanation for this case?

- A. The urine makes the feces dry faster
- B. Uncontaminated urine is excellent fertilizer
- C. Mixing urine with feces retards decomposition
- D. The fertilizer in the feces is better when it is separates

ANSWER: B

6. A municipality of a small town designed a pour flash toilet system of excreta and gray water management to be established for the coming year. During the planning phase, all the factors were assessed and considered to select this system.

What is the most likely primary criterion considered in the selection of this facility?

- A. Durability of the structure
- B. Adequate water supply
- C. Enough space for the system
- D. Affordability of the construction

ANSWER: B

7. After construction of a new septic tank for a small neighborhood community, it was confirmed by the expert that the effluent coming out of the septic tank has a significant number of pollutants that need further attention. Therefore, the expert decided to have additional small-scale treatment for the effluent.

What is the most appropriate next treatment for this case?

- A. Use for irrigation
- B. Sub-surface drain
- C. Secondary septic tank
- D. Discharge to water body

ANSWER: B

8. After CLTS implementation, most members of the densely populated semi-urban community owned pit latrines. The next step was alleviating the fly and odor problem, which was the main concern of the community. Considering the community's good water supply and availability of fecal sludge management service, the sanitation officer promoted pour-flush latrines instead of ventilated improved pit latrines.

What feature of the facility most likely influences the choice in this case?

- A. The vent
- B. The squat
- C. The water seals
- D. The super structure

ANSWER: C

9. In a small growing city, an environmental health expert recommended the use of a sanitation system in which human excreta are processed on-site until they are completely free of disease-causing organisms. In this system, the nutrients contained in the excreta are recycled by using them in agriculture.

What is the most likely type of sanitation technology applied in this case?

- A. Drop and store sanitation system
- B. Pour flush sanitation system
- C. Ecological sanitation system
- D. Centralized sanitation system

ANSWER: C

10. In one of the rural villages of the Horn of Africa, urine and feces were changed to soil reserves and used for plant production. In some cases, the nutrients retrieved were used to restore ecologically degraded non-agricultural lands to create parks and green spaces. By doing so, the use of chemical fertilizers has been significantly decreased and prevented the pollution of aquatic ecosystems.

What is the most likely ecological sanitation approach applied in this case?

- A. Dehydration
- B. Recycling
- C. Reusing
- D. Sanitizing

ANSWER: B

11. The manager of a refugee camp plans to construct sanitation technology that operates without water and has a partition so that urine does not mix with the feces. The main reason for using such technology is to use the urine and dry feces in agriculture. So, the manager consulted the expert to recommend the sanitation technology that is to be constructed in the refugee camp.

What is the most likely type of sanitation technology to be used in this case?

- A. Flush toilet
- B. Shallow trench latrine
- C. Urine-diverting dry toilet
- D. Controlled open defecation

ANSWER: C

12. A sanitation officer was assigned in urban slum, where sanitation status is poor and diarrheal disease is prevalent. The officer has given task to design appropriate sanitation system for the community. The officer has got a grant to construct a sanitation facility at the neighborhood level.

What is the most appropriate sanitation technology that can be used in this setting?

- A. Simple pit latrine
- B. Communal latrine
- C. Ventilated improved pit latrine
- D. Controlled open defecation field

ANSWER: B

13. A pour-flush latrine facility was built for the community living in densely populated slum areas of the country. Due to failure in using appropriate anal cleaning materials, the latrine stopped its function, and the community's health was negatively affected. After a period of time, an international organization discussed with the community about what types of latrine facilities they want to have. All of the community agreed to have a sanitation facility that supports utilization of all types of anal cleaning materials.

What is the most appropriate sanitation system to be recommended for this case?

- A. Ventilated improved pit latrine
- B. Ecological sanitation latrine
- C. Biogas toilet
- D. Simple pit latrine

ANSWER: A

14. In a rural community of Ethiopia, a non-governmental organization built a sanitation system that is aesthetically safe and consistent with the cultural and social values of the community.

What is the most likely criterion of sanitation system addressed in this case?

- A. Affordability
- B. Acceptability
- C. Environmental protection
- D. Disease Prevention

ANSWER: B

15. An environmental health officer asked to recommend appropriate sanitation technology for a densely populated community living in an urban slum area. The area selected to construct the latrine was prone to flooding, and the groundwater table was near to the surface of the earth.

What is the most appropriate sanitation technology to be recommended in this case?

- A. Pour flush latrine
- B. Borehole latrines
- C. Ventilated improved pit latrines
- D. Dehydration vaults

ANSWER: D

16. A nongovernmental organization has installed hand washing facilities to reduce the prevalence of communicable diseases in a sample area. After installing the facilities, training was given for the community on the way of using the facilities. The local government wants to evaluate the equity and inclusion of the project.

What is the most appropriate indicator to be used for this evaluation?

- A. Number of trained households
- B. Number of hands washing facility
- C. Practice of hand washing within the community
- D. Disaggregated monitoring of marginalized groups

ANSWER: D

17. An investigator planned to test the applicability and acceptance of the Urine-Diverting Dry Toilet (UDDT) in the community. As a model, the expert wants to install one Urine-Diverting Dry Toilet (UDDT) to show for the communities.

What is the most critical design consideration for effective performance of the toilet?

- A. Separation of urine and feces
- B. Installation of vertical vent pipe
- C. Pan should have a slope of at least 25 degrees
- D. Installation of silage disposal facility

ANSWER: A

18. A municipality wants to provide a well water supply for a community living around a public latrine. For this purpose, the municipality requested an environmental health expert to identify and recommend an appropriate source of well water with little probability of contamination.

What is the most appropriate healthy water source to be recommended in this case?

- A. Bored
- B. Driven
- C. Drilled
- D. Dug

ANSWER: C

19. A health center asked an environmental health officer to determine the amount of dissolved minerals found from the tap located in the compound. The officer confirmed that the level of fluoride, chloride, iron, and manganese was above the maximum allowable limits.

What is the most likely source of the water coming through the tap?

- A. Well
- B. Rain
- C. Lake
- D. River

ANSWER: A

20. An investigator wants to determine the amount of rain water collected from a corrugated iron roof. The annual rainfall of the area was 80 centimeter, and the area of corrugated iron sheet was 50 square meters.

What is the amount of rain water collected per year in this case?

- A. 140,000 liters
- B. 40,000 liters
- C. 4,000 liters
- D. 40 liters

ANSWER: C

21. A researcher conducted sanitary surveillance of the well water source. The well water was situated downhill from the sanitation facility at a distance of 20 meters. The researcher confirmed that the sanitary risk level of the water source was very high, making the water unacceptable for human consumption.

What is the most likely sanitary risk factor of water supply in this case?

- A. Proximity of the sanitation facility
- B. Location of the water source
- C. Source of the water supply
- D. Soil type of the locality

ANSWER: B

22. A researcher wants to conduct a sanitary survey of surface water sources. The finding of the survey revealed that suspended materials were not adequately removed, and as a result, the water disinfection was not efficient. Further, the finding confirmed that the sanitary risk score of the water supply system was very high, which makes the water unfit for human consumption.

What is the most appropriate control measure to be implemented in this case?

- A. Flotation
- B. Retention
- C. Source protection
- D. Ultraviolet radiation

ANSWER: B

23. An environmental health officer working at a district water and sewerage office presented the sanitary inspection findings of water sources at a national workshop. The report indicated that water is available every day for 24 hours and is located within one kilometer of the residential area. However, there was no water safety plan in this district.

What is the most likely basic service parameter for which the water supply system failed to achieve?

- A. Quality
- B. Continuity
- C. Affordability
- D. Accessibility

ANSWER: A

24. A well water source was developed for the community living in a town. After several periods of time, the water and sewerage authority planned to identify the possible problems and sources of contamination before the water becomes contaminated and the health of the consumer is negatively affected.

What is the most important activity to be performed in this case?

- A. Sanitary inspection
- B. Water quality analysis
- C. Fencing of water sources
- D. Maintenance of pipe lines

ANSWER: A

25. A researcher wants to determine the physicochemical quality of water. The source of the water to be tested varies by quality and quantity throughout the time. As a result, the researcher decided to take samples at known intervals.

What is the most appropriate sampling technique to be applied in this case?

- A. Grab
- B. Composite
- C. Continuous
- D. Flow weighted

ANSWER: B

26. An environmental health officer was requested to determine the bacteriological quality of drinking water supply located in a small town. The source of water has a probability of contamination. The officer wants to check the presence of *E. coli* in the water to determine level of contamination.

What is the most likely reason for selecting *E. coli* as indicator organism in this case?

- A. It is found in cold blooded animals
- B. It represents fecal pathogens
- C. It is disinfection resistant
- D. It is easy to identify

ANSWER: B

27. An environmental health professional is conducting field supervision at ground water sources. During observation, one of the ground waters has brown color.

What is the most likely cause of color change in this case?

- A. The presence of manganese in the water
- B. The presence of iron in the water
- C. The presence of algae in the water
- D. The presence of industrial waste in the water

ANSWER: A

28. An environmental health officer visited drinking water treatment plant. During the visit, the officer noticed that the raw water smells as rotten egg.

What is the most likely cause of odor change in this case?

- A. Ammonia
- B. Hydrogen sulfide
- C. Carbon dioxide
- D. Methane

ANSWER: B

29. An environmental health officer conducted research on bacteriological quality of drinking water. The officer made a serial dilution to get different concentrations, and took 0.1 mL from 0.01 dilution and added onto a plate containing sterilized plate count agar. After incubation at 37 degrees centigrade for 24 hours, the number of colonies found was 10.

What is the most likely microbial count obtained per mL from the original sample?

- A. 1×10 the power of 4
- B. 1×10 the power of 5
- C. 1×10 the power of 6
- D. 1×10 the power of 7

ANSWER: A

30. An environmental health officer planned to determine the suitability of a given water source for domestic purpose. The officer took a sample from raw water and analyzed for five days Biochemical Oxygen Demand (BOD₅). After the investigation, the BOD₅ was determined to be 3.5 parts per million (ppm).

What is the most likely explanation for the finding in this case?

- A. Poor source of water supply
- B. Moderate source of water supply
- C. Good source of water supply
- D. Excellent source of water supply

ANSWER: A

31. An international organization, which has a plan to construct small-scale water treatment technology for a community, consulted an environmental health officer to identify appropriate water treatment technology selection criteria. The organization will only give adequate training for the community and recruit an expert to manage the overall process of the treatment technology.

What is the most appropriate selection criterion used in this case?

- A. Economy
- B. Type of water
- C. Awareness of the community
- D. Presence of skilled man power

ANSWER: A

32. A particular household moved to a temporary settlement area where a rural water supply system is absent. As a result, it was forced to use a household water treatment technique. Although the water is treated and has a flat taste, the children were not willing to drink it.

What is the most likely water treatment technique used by the household in this case?

- A. Boiling
- B. Filtration
- C. Chlorination
- D. Ultraviolet radiation

ANSWER: A

33. A researcher wants to determine the optimum dose of a coagulant to achieve the required level of suspended solid removal from surface water. To this effect, four settling equipment labeled as A, B, C, and D, each filled with 1 liter of surface water sample, were prepared. The initial turbidity level of water in each piece of settling equipment was reported to be 30, 50, 80, and 100 Nephelometric Turbidity Units (NTU), respectively. Then, the researcher added 10, 20, 30, and 40 mg of the coagulant into the settling equipment, respectively, and shook vigorously with a glass rod. The samples were left for 30 minutes to flocculate and allowed to stand in quiescent conditions for 60 minutes. The final turbidity level was reported to be 20, 45, 60, and 85 NTU, respectively.

What dose of a coagulant that gives better turbidity removal in this case?

- A. 10 mg
- B. 20 mg
- C. 30 mg
- D. 40 mg

ANSWER: A

34. A water source was constructed for primary school. The students use this water source for drinking and personal hygiene purposes. Once upon a time, some of the students complained about the occurrence of health problems. The school administration, in collaboration with the town health department, collected stool samples from the students and confirmed that students were positive for *Entamoeba histolytica*.

What is the most appropriate filtration type to be used for removing this parasite?

- A. Diatomaceous earth filter
- B. Rapid sand filter
- C. Slow sand filter
- D. Gravity filter

ANSWER: C

35. Water pipes and fixtures of a given water distribution system were corroded. After taking a sample from the water and analyzing for the presence of gases, it was confirmed that excess carbon dioxide and hydrogen sulfide were the major cause of corrosion of pipes and fixtures.

What is the most effective method for removing carbon dioxide and hydrogen sulfide in this case?

- A. Aeration
- B. Coagulation
- C. Filtration
- D. Sedimentation

ANSWER: A

36. A new water well has just been constructed for a small community in a rural village and has to be disinfected before use. An environmental health officer wanted to prepare a chlorine stock solution. The total daily water yield of the well is 2000 liters. The water has low turbidity and requires to be treated with 1 ppm of chlorine. Chlorinated lime with 30 percent chlorine is available in the local market.

What is the most appropriate daily chlorine requirement to prepare the stock solution?

- A. 3.6gm
- B. 4.6gm
- C. 5.6gm
- D. 6.6gm

ANSWER: D

37. The laboratory technician in the water treatment plant measured the residual chlorine after disinfection. The result showed that combined residual chlorine is below the minimum standards. When the experts investigated the disinfection process, 0.5 mg/L chlorine was added, the contact time was 40 minutes, the pH was 6-8, and the temperature was at room temperature.

What is the most likely factor that lowers the residual chlorine in this case?

- A. Contact time
- B. H ion concentration
- C. Chlorine concentration
- D. Temperature of the water

ANSWER: C

38. Water-associated disease is a prevalent problem in a community. As an intervention to this problem, the municipality together with other stakeholders worked to achieve a target of doubling the proportion of households with piped water supply.

What is the most appropriate water service indicator that needs to be monitored for this case?

- A. Accessibility
- B. Affordability
- C. Continuity
- D. Quantity

ANSWER: A

39. A community living in tropical area, uses river water and is frequently affected by waterborne disease. The public health officer wanted to recommend home based water treatment. The officer took water sample and identified different forms of microorganism including cyst and spore forming organisms.

What is the most efficient home-based water treatment recommended for this case?

- A. Boiling
- B. Chemical disinfection
- C. Filtration
- D. Solar disinfection

ANSWER: A

40. A community living in downstream of the city complained about the pollution of the nearby river with wastewater discharges from the city. The city administration conducted river water quality analysis and found that the river contains millions of bacteria and virus that may negatively affect the health of the community.

What is the most likely source of river water pollution in this case?

- A. Domestic
- B. Industrial
- C. Institutional
- D. Storm

ANSWER: A

41. A city administration constructed the closed type of wastewater sewerage system to convey waste water to the plant which was initially designed to treat 8000-meter cube of waste water in a day. However, its efficiency declined since the treatment plant receives wastewater higher than the daily treatment capacity.

What is the most likely source of additional waste water flowing to this treatment plant?

- A. Domestic
- B. Industrial
- C. Infiltration
- D. Storm

ANSWER: C

42. An investigator planned to measure the average daily flow, maximum daily flow, peak hourly flow, minimum hourly and daily flows, and design peak flow from a given wastewater treatment plant. The aim of this initiative was to know the generation rate for the collection and interceptor sewers, wet wells, wastewater pumping stations, and waste-water flow measurements.

What is the most appropriate flow rate factor used in this case?

- A. Average daily
- B. Maximum daily
- C. Maximum hourly
- D. Minimum hourly

ANSWER: C

43. The environmental health expert working in Zonal office wants to estimate the hourly capacity of sewerage system in a given University. There are 10000 students in the University and each student produces 60 liters of wastewater per day.

What is the most likely wastewater flow rate in meter cube per hour for this case?

- A. 25
- B. 50
- C. 100
- D. 150

ANSWER: A

44. Communities living in the downstream of the wastewater treatment plant complained about pollution of the water source that receives effluent from the treatment plant. Based on the complaint, the investigator decided to determine the average concentration of pollutant in mass per unit time.

What is the most appropriate sampling method to be used in this case?

- A. Grab
- B. Direct
- C. Composite
- D. Continuous

ANSWER: C

45. The environmental health officer was requested to determine the level of color of a given wastewater treatment plant. Since the treatment plant was very far from the laboratory where the analysis is performed, the officer is unable to start sample analysis timely and decided to refrigerate the sample in dark area.

What is the maximum refrigeration time of the sample in dark area for this case?

- A. 1 day
- B. 2 days
- C. 3 days
- D. 4 days

ANSWER: B

46. The town administration wants to construct an activated sludge wastewater treatment plant with ultraviolet radiation as disinfection options. However, the disinfection process was not effective due to the presence of interferences in waste water stream.

What is the most likely parameter that interferes with the disinfection process in this case?

- A. Color
- B. Odor
- C. Temperature
- D. Turbidity

ANSWER: D

47. Communities who have private septic tank asked the municipality to have an adequate wastewater collection system. Based on the request, the municipality discussed with the communities and agreed to construct suitable wastewater collection system.

What is the most appropriate wastewater collection system for this case?

- A. Advanced sewer
- B. Conventional sewer
- C. Simplified sewer
- D. Solid free sewer

ANSWER: D

48. A city administration consulted the environmental health expert to select a suitable wastewater collection system for a densely populated city, which collect and transport the wastewater to the designed central treatment. The wastewaters are generated from residences, businesses, and some industrial activities.

What is the most appropriate sewer collection type for this case?

- A. Sanitary
- B. Combined
- C. Storm
- D. Industrial

ANSWER: A

49. The environmental health experts are investigating the choice of wastewater treatment technologies for a city that generated a large quantity of wastewater. The experts reviewed the contaminant removal efficiency of each of the available technologies by considering various quality parameters. Finally, they decided to implement treatment technology called the Anaerobic Baffled Reactor.

What is the most important parameter considered by the experts?

- A. Dissolved oxygen
- B. Biochemical oxygen demand
- C. Nutrients
- D. Suspended solid

ANSWER: B

50. The municipality of a town wants to construct appropriate wastewater treatment technology for a given community. The municipality planned to reduce the level of solid by half and Biochemical Oxygen Demand (BOD) by 30 to 40 percent.

What is the most appropriate treatment technology for this case?

- A. Septic tank
- B. Lagoon
- C. Constructed wetland
- D. Soak pit

ANSWER: A

51. A municipality planned to construct a wastewater treatment plant for the city. For this purpose, the amount and constituents of wastewater, capacity, efficiency, reuse potential, complexity, and the cost of each of the possible treatment plant was studied. Based on the finding, the municipality decided to construct advanced wastewater treatment technology.

What is the most appropriate criterion to select advanced treatment technology?

- A. Complexity of the technology
- B. Reuse potential
- C. Cost of the technology
- D. Quantity of the generated wastewater

ANSWER: B

52. A researcher wants to determine the various solid types of a given treatment plant. The researcher found suspended solids of greater than 1micrometers, dissolved solid of less than 1micrometers, soluble Biochemical Oxygen Demand (BOD) of less than 1micrometers and BOD in suspension form with a size of greater than 1micrometers.

What is the most appropriate operation unit to control soluble BOD?

- A. Filtration
- B. Screening
- C. Sedimentation
- D. Stabilization

ANSWER: D

53. An environmental expert working in a wastewater treatment plant wants to measure the suspended solid removal efficiency of the treatment plant. The concentration of the suspended solid in the influent was 500 milligram per liter. After the wastewater treated in primary and secondary treatment processes, it was found to be 50 milligram per liter in the effluent.

What is the most likely percentage of the suspended solid removal efficiency of the treatment plant?

- A. 10
- B. 90
- C. 50
- D. 450

ANSWER: B

54. A researcher planned to measure the municipal solid waste generation rate in a given town. Food wastes, paper waste, cardboard, plastic, textile, leather, yard waste, wood, glass and tin were the major physical composition of the waste.

What is the most likely source of solid waste included in this case?

- A. Agricultural waste
- B. Residential waste
- C. Industrial process waste
- D. Treatment plant site waste

ANSWER: B

55. A researcher planned to measure a waste generation rate in a given town. For this purpose, plastic bags were distributed for the households to collect solid waste for seven consecutive days. In each day, the amount of solid waste generated from the households was measured. About 130kg, 90kg, 100kg, 115kg, 120kg, 110kg, 125kg of waste was measured on the 1st, 2nd, 3rd, 4th, 5th, 6th, and 7th days respectively. The total population of the town was 950.

What is the most likely total domestic solid waste generation rate in this case?

- A. 0.118 kg/c/d
- B. 0.75 kg/c/d
- C. 0.83 kg/c/d
- D. 5.82 kg/c/d

ANSWER: C

56. A researcher wants to determine the characteristics of a given hazardous waste stream. After performing the required analysis on laboratory animal, it is confirmed that the waste has a potential to cause mutagenic and teratogenic effect.

What is the most likely characteristic of the waste in this case?

- A. Toxic
- B. Reactive
- C. Ignitable
- D. Corrosive

ANSWER: A

57. A health care professional working in a hospital generated wastes including dressings, bandages, gauze, cotton, gloves and masks contaminated with blood and body fluids; while giving healthcare services for a patient in emergency ward.

What is the most likely type of hazardous waste generated in this case?

- A. Chemical waste
- B. Anatomical waste
- C. Radioactive waste
- D. Infectious waste

ANSWER: D

58. A laboratory technician working in a hospital generated wastes including reagents, fixatives, formalin, xylene, toluene, methanol, methylene chloride and other solvents while conducting a laboratory procedure for a patient.

What is the most likely type of hazardous waste generated in this case?

- A. Infectious waste
- B. Chemical waste
- C. Radioactive waste
- D. Pharmaceutical waste

ANSWER: B

59. In a given city the solid waste generation rate is increasing from time to time, and the mayor of the city assigned a researcher to identify the cause. The researcher conducted an assessment and confirmed that the fertility rate is constant and there was no urban expansion. The researcher also confirmed that the community's living standards and culture are also unchanged; however, the national Gross Domestic Product (GDP) has increased.

What is the most important factor affecting waste generation rate in this case?

- A. Urbanization
- B. Public habits
- C. Economic growth
- D. Population growth

ANSWER: C

60. In a given town the solid waste collectors locally named 'koralio' were collecting the broken glasses, metals and plastic shoes from households and sell it to companies that changes these waste materials in to new ones.

What is the most appropriate waste diversion method used in this case?

- A. Sorting
- B. Reusing
- C. Recycling
- D. Recovery

ANSWER: C

61. A hospital was disinfecting the floor surface of the wards using chemicals. Meanwhile, the environmental health officer conducted risk assessment and identified that these chemicals may cause health problem for cleaners and inform the manager to change the chemical disinfection to steam disinfection instead.

What is the most appropriate waste minimization method applied in this case?

- A. Reuse
- B. Recycle
- C. Recovery
- D. Source reduction

ANSWER: D

62. Environmental health officer of a town conducted a waste characterization study and found a high proportion of organic waste components that have very low moisture content and ash.

What is the most appropriate waste diversion method used in this case?

- A. Recycling
- B. Composting
- C. Anaerobic digestion
- D. Waste-to-Energy Incineration

ANSWER: D

63. A municipality planned to construct a waste recycling facility. It is found that the non-recyclable materials have not been completely separated out from the waste stream which may challenge the implementation of recycling process.

What is the most likely factor affecting the recycling process in this case?

- A. Quality of the waste
- B. Contamination of the waste
- C. Quantity of the waste
- D. Lack of skilled man power

ANSWER: B

64. A company is manufacturing electronic products and distributes the products to the users by opening the distribution centers. Besides, it developed a system of collecting old and used products by asking customers to pay birr when purchasing the product and reimbursing the paid birr when returning the product for recycling and safe disposal.

What is the most appropriate principle used to collect waste in this case?

- A. Deposit-refund scheme
- B. Advance disposal fee
- C. Material tax
- D. Duty of care

ANSWER: A

65. A municipality of a town developed a solid waste collection system for a residential area by putting large waste storage containers at different places in the community. When a container is full, the municipality sends a waste collection vehicle to pick up the container and haul it in a disposal place and replace the empty container by another container.

What is the most likely waste collection system used in this case?

- A. Block collection system
- B. Curb side collection system
- C. Communal collection system
- D. Door to door collection system

ANSWER: C

66. A municipality has no communal container to collect solid waste generated from households. As a result, the inhabitants dispose solid waste to the surrounding environment. The municipality decided to establish a collection system that involves picking the waste loaded container and emptying the content into the vehicle and transport the waste to the disposal facility so that the waste storage container remains at point of generation for subsequent use.

What is most appropriate solid waste collection system used in this case?

- A. Hauled container system
- B. Stationary container system
- C. Curbside collection system
- D. Communal collection system

ANSWER: B

67. A hotel generates a huge amount of garbage waste, and the manager of the hotel ordered the purchasing officer to buy proper waste containers to store the waste.

What is the most appropriate type of container recommended in this case?

- A. Pushcart
- B. Plastic bag
- C. Perforated plastic bin
- D. Galvanized metal

ANSWER: D

68. A surgeon working in a hospital generates liquid infectious waste while conducting a surgical procedure for a patient in an operation theater ward.

What is the most appropriate hazardous waste treatment method to be used in this case?

- A. Autoclave
- B. Incineration
- C. Microwave treatment
- D. Chemical disinfection

ANSWER: D

69. A hotel generates huge amount of food leftover, yard and tree trimmings, and the manager of the hotel consults the environmental health officer of the town to recommend an appropriate solid waste management system.

What is the most appropriate solid waste management system for this scenario?

- A. Pyrolysis
- B. Hog feeding
- C. Composting
- D. Onsite incineration

ANSWER: C

70. A municipality planned to construct a landfill to dispose solid waste in sanitary way and consulted an environmental health officer to recommend appropriate site for land fill construction. The proposed sanitary landfill should minimize the potential for flooding and contamination of water bodies.

What is the most appropriate site selection criterion to be used in this case?

- A. Potential landfill capacity
- B. Distance from surface water bodies
- C. Geological and hydrological elements
- D. Number of populations served

ANSWER: B

71. A training center provides capacity-building training for farmers about composting process and available technologies. After completing the training, the farmers collect only green waste and start to prepare compost. However, the composting process produces undesirable odors irritant for the farmers themselves and the surrounding community.

What is the most likely factor responsible for creating undesirable odors in this case?

- A. PH
- B. Carbon to Nitrogen ratio
- C. Bulk density
- D. Feedstock particle size

ANSWER: B

72. A team from environmental protection authority conducted a study and found that chemical leached from landfill sites cause groundwater contamination. Besides, an overflow of pit latrines and latrine facility damage was observed. Frequent death of animals was also creating handling of carcasses problematic; as a result, extreme weather conditions occurred in the study area.

What is the most likely environmental health service affected in this case?

- A. Health Surveillance of premises
- B. Prevention of communicable diseases
- C. Food quality control
- D. Waste management

ANSWER: D

73. In a town, heavy rainfall season that persisted for two months was prevalent. The rainfall damaged the available environmental health infrastructures and the flooding destroyed different habitats in the town, and created stagnant water. An environmental health team has conducted institutional survey to determine the magnitude of environmental health related diseases in all public and private health facilities.

What is the LEAST LIKELY disease to be reported during the two months high rainfall season in the town?

- A. Water borne diseases
- B. Sanitation related diseases
- C. Vector borne diseases
- D. Food borne diseases

ANSWER: C

74. The environmental protection authority reported that a brownish smoke has been formed on the sky. Besides, there was reduced visibility in the city and plants were also damaged. The city residents were also encountering eye, nose and throat irritation.

What is the most likely atmospheric or meteorological consequence of air pollution in this case?

- A. Acid deposition
- B. Global warming
- C. Photochemical smog
- D. Ozone depletion

ANSWER: C

75. A study conducted by a group of researchers to identify the effect of climate change on meteorological events indicated that some parts of the globe have experienced high amount of rainfall and other places experienced drought.

What is the most likely air pollutant responsible for this observed variation?

- A. Sulfur dioxide
- B. Carbon dioxide
- C. Carbon tetrachloride
- D. Nitrogen dioxide

ANSWER: B

76. A study conducted in an industrialized country regarding the effect of air pollutants on human health. And, the findings revealed respiratory irritation, bronchitis and mucus production among the exposed population. Besides, skin redness, damage to the eyes like lacrimation, corneal opacity and mucous membranes were observed among many of the exposed people.

What is the most likely air pollutant causing the problem in this case?

- A. Sulfur dioxide
- B. Nitrous Oxide
- C. Silicon dioxide
- D. Carbon dioxide

ANSWER: A

77. An abandoned sanitary landfill in the borders of a city was found to be the major cause of global warming, according to the study conducted by environmental protection authority (EPA). The EPA measured global warming indicator pollutants emitted from the sanitary landfill to support their conclusion.

What is the most likely indicator used by the EPA in this case?

- A. Methane
- B. Nitrous oxide
- C. Chlorofluorocarbon
- D. Carbon dioxide

ANSWER: A

78. Condominium residents were growing vegetables in their available small yard using the nearby river water for irrigation. The environmental health officers have identified the open defecation was very common in the upstream area. Hence, the officers claimed the vegetables grown might cause health problems and planned to take vegetable samples and verify their concern.

What is the most appropriate parameter to be used in this case?

- A. Microbial
- B. Chemical
- C. Physical
- D. Invertebrates

ANSWER: A

79. Untreated wastewater was flowing from a hospital to the nearby river. The local environmental health officer planned to conduct river quality assessment aimed at determining the pollution impact of the hospital wastewater. To this effect, the officer applied bio-monitoring protocol in the study.

What is the most appropriate bio-indicator to be used in this case?

- A. Phytoplankton
- B. Zooplankton
- C. Macrophyte
- D. Macroinvertebrate

ANSWER: D

80. A river crossing through agricultural lands and a big city in one of the developing countries receives organic pollutants from the agricultural land and untreated wastewater discharge from the city. To determine the potential impacts of these wastes on the river ecosystem, an environmental health expert planned to study the number of organic wastes coming from these pollution sources.

What is the most important water quality parameter to be used in this case?

- A. Biochemical oxygen demand
- B. Volatile organic carbon
- C. Polyaromatic hydrocarbon
- D. Organic pesticides

ANSWER: A

81. An ecologist planned to study the effect of turbidity on the light penetration capacity and aquatic life forms in a lake found in East Africa. To this end, the ecologist measured the turbidity of water using turbidometer and the depth of light penetration using Secchi disc.

What is the most appropriate bio- to be used for attaining the objective of this study?

- A. Nekton
- B. Phytoplankton
- C. Zooplankton
- D. Periphyton

ANSWER: B

82. A team of environmental health professionals working in a national park conducted a study on the distribution of lion species. Based on their assessment, they found the lions have such characteristics as living and hunting together and they are mainly found in places where herds of Wild Beast and Impala are present.

What is the most likely population distribution pattern indicated in this scenario?

- A. Uniform
- B. Random
- C. Regular
- D. Clumped

ANSWER: D

83. Due to the expansion of urbanization, the vegetated land cover has been converted into hardened surfaces in one of the cities in Ethiopia. As a result, the temperature of the city and the surrounding area has been affected.

What is the most likely ecosystem services affected in this city?

- A. Provisioning service
- B. Regulation service
- C. Cultural service
- D. Supporting service

ANSWER: B

84. An environmental protection authority (EPA) encouraged the major industries in the city to use solar, hydroelectric, and geothermal energy to reduce their dependency on fossil fuel.

What is the most likely pollution control method encouraged by EPA?

- A. Source control
- B. Dilution
- C. Collection of pollutants
- D. Treatment

ANSWER: A

85. A company engaged in production of paper coffee cup has conducted a study to decide on the possibility of future production. The study collected and measured the data on the number of raw materials used, the production process, the transportation, distribution, use, recycling potential as well as final disposal of the cup. Based on the findings, the company decided to halt the production by citing potential environmental impact of the coffee cup.

What is the most likely pollution management system applied by the company in this case?

- A. Life cycle assessment
- B. Environmental management system
- C. Environmental risk assessment
- D. Environmental impact assessment

ANSWER: A

86. A team of environmental health experts wants to treat a portion of soil contaminated by hazardous hydrocarbon pollutants. The team primarily performed site investigation and found that the area has enough amounts of nutrients but with low microbial load that can degrade the pollutants.

What is the most likely human intervention applied for enhancing bioremediation in this case?

- A. Bio-augmentation
- B. Bio-stimulation
- C. Bio-venting
- D. Bio-sparging

ANSWER: A

87. A study conducted in the sea showed alive sea turtles were more worth to local communities than the dead ones. The study team gathered the local community and advised them to engage more on tourism than selling the eggs and meat of sea turtle. Hence, the biodiversity of the sea will be protected and the local community will be more beneficial.

What is the most likely aquatic biodiversity protection method applied in this case?

- A. Polluter-pay-principle
- B. Economic incentive
- C. Protection of sea sanctuary
- D. Community commitment

ANSWER: B

88. Acid mine drainage from abandoned mine polluted the nearby lake, which is providing an ecosystem service to the surrounding community. The local environmental protection authority grows plants and allows the drainage to pass through it before the drainage enters to the lake. The plant roots immobilize the metals by their roots. The plants also produce organic matter to the area that further enhances metal absorption by organic matter.

What is the most likely pollution control method applied in this case?

- A. Activated sludge system
- B. Oxidation pond
- C. Constructed wetland
- D. Trickling filter

ANSWER: C

89. In one of the cities of the country, there was frequent conflict. Due to this, the authority of the government planned to resettle the community in new area which is conflict free. In the new settlement area, there is church or mosque, playgrounds for children, markets, kebele and police offices, and recreational sites for the community. Most of the community members feel happy.

What is the most likely principle of healthful housing fulfilled in this case?

- A. Protection of physiological needs
- B. Protection against accidental injuries
- C. Protection against psychological stress
- D. Protection against communicable diseases

ANSWER: C

90. In residential area of well-planned city adequate water is available for proper utilization of latrines, proper solid waste storage and disposal, as well as proper hand washing facilities.

What is the most likely principle of healthful housing fulfilled in this case?

- A. Protection against Physiological needs
- B. Protection against excessive noise
- C. Protection against accidental Injuries
- D. Protection against communicable diseases

ANSWER: D

91. A house located in slum area of a city is characterized by lack of windows for light and ventilation, as well as overcrowding of sleeping rooms. The infrastructure of the house is also not well maintained. Due to this, there is water leakage. Every morning, the wall of the house is moist and mold growth is manifested.

What is the most likely disease to prevail in this case?

- A. Asthma
- B. Diarrhea
- C. Scabies
- D. Typhus fever

ANSWER: A

92. An environmental health expert, investigating the quality of the houses in a given residential area, observed that the house is made from concrete and cements, the walls are well plastered, and the rooms are well ventilated. However, when the expert tried to record the result on the checklists, it was not visible inside the house. So, the expert was forced to go out of the house to record the result.

What is the most likely factor that affected the quality of the house in this case?

- A. Illumination
- B. Ventilation
- C. Air conditioner
- D. Structure of the wall

ANSWER: A

93. A house located in slum area of the city were poorly sited, planned and constructed. There is lack of basic sanitation services. The occurrence of communicable disease is high in this area. The local government plans to reestablish the community in new area.

What is the most appropriate criterion to be used for selecting residential district in this case?

- A. Accessibility of the road
- B. Accessibility of construction materials
- C. Accessibility of electricity
- D. Accessibility of drinking water

ANSWER: D

94. A government plan to build an elementary school in the urban village. Before building the school a team of experts conduct survey analysis, and the village is known by floods.

What is the most appropriate school site selection criterion should be considered?

- A. School site should be centrally located
- B. School site should be elevated
- C. School site should be 1 km away from marshy areas
- D. School site should have sufficient playground

ANSWER: B

95. A family living in rural area of the country uses charcoal to get heat at the home during summer season. In one night, while using this charcoal, one of the family members felt muscles weakness, headache, dizziness, nausea, vomiting, psychomotor impairment, dimness of vision, and coronary effects.

What is the most likely cause for the observed problems in this family?

- A. Combustion products
- B. Building chemicals
- C. Biological aerosols
- D. House hold dusts

ANSWER: A

96. A diarrhea outbreak has repeatedly occurred in prisons. An environmental health expert visited one of the prisons to identify the possible factors for the occurrence of the outbreak. During the visit, the expert observed that the prison is located at downhill, rooms do not get natural and artificial light as well as adequate ventilation, and there is no enough facility to keep sanitation and hygiene.

What is the most likely factor that affects the health condition of the prisoners in this case?

- A. Scarcity of water
- B. Location of the prison
- C. Illumination
- D. Ventilation

ANSWER: A

97. A high school, which is found in rural village of the community, has a total number of 500 students out of which almost half of them are females. The school has only two separate toilets for males and females. Most of female students are absent during menstruation period.

What is the most likely health services component that should be improved in this school?

- A. Access to drinking water
- B. Adequate sanitation facilities
- C. Cleanliness of the school
- D. Proper ventilation

ANSWER: B

98. An environmental health professional works as a safety officer in a textile industry. The industry has a high level of dust emission that varies by working section. The officer planned to take administrative measures to reduce dust exposure but the industry allocated limited resource for safety intervention.

What is the most likely administrative intervention that is applicable for this industry?

- A. Safety training
- B. Job rotation
- C. Rest breaks
- D. PPE supply

ANSWER: B

99. An industry was established to produce high quality products. The industrial manager wants to assure the safety and health of the workers and its environment. As a result, an environmental health expert was assigned to inspect the overall safety practice of the industry. The assessment result indicated that the industry applies the least advised hazard control method.

What is the most likely control method to be used in this case?

- A. Personal protective equipment
- B. Engineering Control Methods
- C. Administrative Control Methods
- D. Substitution another resource

ANSWER: A

100. An environmental health officer is assigned in an iron and steel factory as safety officer. The workers complain the presence of excessive heat stress. The company owner was worried about the heat stress and reduced productivity of the worker. As a result, the officer was assigned to ensure the health and safety of the workers. Consequently, strategies to reduce heat stress were designed and implemented.

What is the most appropriate prevention strategy to be used in this case?

- A. Increase work demand
- B. Increase air temperature
- C. Increase air movement
- D. Increase radiant heat

ANSWER: C

101. A safety and health officer are assigned to establish a new wood industry that produces particleboard. Prior to the commencement of the project, the officer has reviewed similar industry having risk of chemical and physical hazards.

What is the most appropriate term describing this case?

- A. Hazard recognition
- B. Hazard anticipation
- C. Hazard evaluation
- D. Hazard control

ANSWER: B

102. An environmental health officer is working in a newly established tannery. Prior to the commencement of the tannery, the officer put hazard mitigation mechanism in order to promote health and safety of the workers.

What is the most appropriate plan of action describing this case?

- A. Hazard recognition
- B. Hazard anticipation
- C. Hazard evaluation
- D. Hazard control

ANSWER: D

103. A safety expert is engaged in an industry ensuring the safety of the workers in tannery industry. For effective hazard prevention, the expert wants to know type of hazards available in the industry. However, there is no documentation that can give a clue about it.

What is the most likely method of assessment that the safety expert can use in this case?

- A. Audit report analysis
- B. Workplace inspection
- C. Reviewing ill health statistics
- D. Investigation of accident records

ANSWER: B

104. A workers of pesticide formulation industry experienced physical hazards due to their malpractices. The company owner appointed a safety expert to make a risk assessment and the expert generated a report based on evidences gathered.

What is the most primary target for risk assessment in this case?

- A. Professional associations
- B. Employer
- C. Employee
- D. Researcher

ANSWER: C

105. A worker in a company is often engaged in a communication room to reach out remotely located branch offices which emit radiation. Upon long time exposure, the worker developed a cataract. The safety expert made a hazard assessment to identify the type of radiation the worker is exposed. What is the most likely type of radiation exposure in this case?

- A. UV Radiation
- B. Gama Radiation
- C. Laser Radiation
- D. Microwave Radiation

ANSWER: C

106. An environmental health expert working in an automotive industrial process is concerned about the mechanism of enhancing the health and safety of the workers by minimizing ergonomic hazards. The expert implemented capacity building coupled with supply of safety materials. Afterwards, the expert wants to evaluate the impact of the interventions made.

What is the most appropriate outcome indicator to be used in this case?

- A. Reduced productivity
- B. Reduced efficiency
- C. Reduced income
- D. Reduced injury

ANSWER: D

107. In an industry there was an imbalance between the work demand and capacity of the workers. The health and safety expert made an assessment and identified problems. Based on the assessment, the expert reported sleeping problem, change in eating habits, increased smoking and drinking habit, avoidance of friends and family and sexual problems among the workers.

What is the most likely category of symptom best described in this case?

- A. Emotional changes
- B. Mental changes
- C. Physical symptoms
- D. Behavioral changes

ANSWER: D

108. The productivity of workers in a given company declined from time to time that worried the manager. An environmental health officer conducts an assessment and found the potential work-related problem in the workplace. The main cause of the problem was that some selected workers were asked to increase the production of a newly introduced product than expected before.

What is the most appropriate determinant factor identified in this case?

- A. Homogeneity of tasks
- B. Heterogeneity of tasks
- C. Responsibility of workers
- D. Imbalanced work demand

ANSWER: D

109. A flower farm company produces fresh-cut flowers for the international market. The company is one of the top sources of income for the country. However, in the process, there are toxic chemicals used at the expense of health of the workers and nearby environment. An environmental health expert was assigned as safety expert to suggest possible hazard control measure. However, after evaluation, the best approach was suggested but was found to be not practical.

What is the primary and best approach suggested by the expert for this case?

- A. Substitution
- B. Elimination
- C. Isolation
- D. Enclosure

ANSWER: B

110. Community members living close to the mining industries, small-scale smelter sites, and natural outcroppings complained about the possible health impacts they encounter.

What is the most likely source of the environmental toxicant in this case?

- A. Pesticides
- B. Heavy metals
- C. Mycotoxins
- D. Allergen

ANSWER: B

111. A study revealed that dwellers, mostly children and elders, living in a household with cracks of floor and walls were at higher risk of developing lung cancer compared to similar population groups with standard households.

What is the most likely source of toxicant that causes the lung cancer in this case?

- A. Exposure to tobacco smoking
- B. Exposure to ambient air pollution
- C. Exposure to radon gas
- D. Exposure to dietary source of chemicals

ANSWER: C

112. An environmental health professional has been interested to characterize the toxicological profile of newly introduced chemical. The expert used laboratory animals and exposed them to varied concentration level of the chemical. Based on the experiment, the professional determined the concentration at which there is no adverse effect on exposed population groups.

What is the most likely concentration level the professional identified in this case?

- A. Lethal concentration
- B. Effective concentration
- C. Threshold concentration
- D. Lethal dose concentration

ANSWER: C

113. Workers in a plastic industry accidentally exposed for cadmium, which is in the industry as a pigment and developed nausea, vomiting, and abdominal pain.

What is the most likely route of exposure for cadmium in this case?

- A. Inhalation
- B. Ingestion
- C. Dermal
- D. Intra-peritoneal

ANSWER: B

114. Sanitation workers who are working in the manholes, which are enclosed spaces, sometimes experienced unconsciousness.

What is the most likely environmental hazard the sanitation workers faced in this case?

- A. Fire and explosion
- B. Radiation hazard
- C. Oxygen deficiency
- D. Chemical's exposure

ANSWER: C

115. Housekeeping workers in a hotel work for prolonged time and engaged in different activities like, bed making, cleaning the rooms, etc. Hence, mostly these workers developed a health problem that affects their musculoskeletal system.

What is the most likely hazard the workers are exposed in this case?

- A. Chemical
- B. Physical
- C. Ergonomic
- D. Biological

ANSWER: C

116. An Environmental Health Officer wants to conduct risk assessment on the child toys imported from abroad. The officer took samples from toys and five heavy metals were detected in the laboratory.

What is most likely risk assessment stepping the officer applied in this case?

- A. Exposure assessment
- B. Hazard identification
- C. Risk communication
- D. Risk characterization

ANSWER: B

117. A team from Ethiopian Food and Drug Authority (EFDA) took fish samples and measured the concentration of methyl mercury (HgCH_3). Then the team compared the average concentration of the chemical with the standard value. The standard value was found to be 15 percent higher than the measured value.

What is the most likely risk level of the methyl mercury (HgCH_3) in this case?

- A. High risk level
- B. Low risk level
- C. Medium risk level
- D. Potential risk level

ANSWER: B

118. A comprehensive hospital has been disposing chemicals for the past 20 years. The community residing close to the disposal site has encountered health problems. Children who have been playing around the disposal site were highly affected. Due to the community complaint, the health authority-initiated action to reduce the community sufferings, at least for the future.

What is the appropriate risk management option the health authority could apply in this case?

- A. Providing monetary compensation
- B. Revoking the license of the hospital
- C. Exposure protection
- D. Arranging free health service

ANSWER: C

119. A toxicologist has conducted dietary risk assessment of mercury. The risk assessment finding indicated that the population was at higher risk of mercury poisoning. To reduce the risk level, the toxicologists recommended the community to reduce fish consumption. The toxicologist intended to know the recommendation given was effective to reduce the risk level or not.

What is the most likely risk management option to be applied in this case?

- A. Developing a monitoring plan
- B. Developing exposure assessment plan
- C. Developing risk characterization plan
- D. Developing risk analysis plan

ANSWER: A

120. An environmental health officer who is working in the regional health bureau conducted surveillance by using data from weekly and monthly report that was sent from different health care facilities to provide scientific baseline data for planning, implementing, and evaluating disease prevention and control program.

What is the most likely type of surveillance used in this case?

- A. Active
- B. Passive
- C. Sentinel
- D. Progressive

ANSWER: B

121. A Woreda with high prevalence of trachoma among school children was reported. An Environmental Health officer visited multiple schools and identified that the surrounding area of one of the schools was full of solid wastes and common breeding site for mechanical vectors, which was suspected as the main causes of trachoma transmission.

What is the most likely vector that can transmit the specified disease in this case?

- A. Mosquitoes
- B. House flies
- C. Sand flies
- D. Tsetse flies

ANSWER: B

122. In March 2023, the woreda health office reported 100 cases of relapsing fever, among which 60 of them were new cases. The average total population of the Wereda was 6000.

What is the incidence rate of relapsing fever in the woreda on March, 2023 in this case?

- A. 10 new cases per 1000 population
- B. 20 new cases per 1000 population
- C. 100 new cases per 1000 population
- D. 200 new cases per 1000 population

ANSWER: A

123. The regional health bureau formed a team of health professionals to conduct a survey in cholera endemic area. The team found that the prevalence of cholera in one Kebele, in a year 2023 was 25 percent. They did examination for all residents of the Kebele.

What is the most likely explanation for this finding?

- A. 25 percent of the residences were newly affected by cholera.
- B. There was a total of 25 new cholera cases in kebele.
- C. Only 25 percent of the populations are at risk of cholera infection.
- D. 25 percent of the population was affected by cholera.

ANSWER: D

124. An environmental health officer who is working in regional health bureau observed an excessive number of cases than usual frequency on weekly report of the health center. Based on the report, an epidemic investigation team traveled to a health center to verify the existence of an epidemic.

What is the most likely method of epidemic detection applied in this case?

- A. Data from routine surveillance
- B. Data from community survey
- C. Report from clinicians
- D. Report from the community

ANSWER: A

125. According to WHO report, most of the children in developing countries are highly affected by diarrhea, WHO also stated that diarrhea occurred persistently in high level throughout the year among these countries.

What is the most appropriate explanation for the level of the disease occurrence in this case?

- A. Epidemic
- B. Hyper endemic
- C. Sporadic
- D. Pandemic

ANSWER: B

126. A 40-year-old male patient visited the health center for the wound care and treatment of dog bite wound. The laboratory investigation result confirmed that the patient developed rabies, and he must start treatment immediately.

What is the most likely route of disease transmission for this case?

- A. Direct contact
- B. Direct projection
- C. Indirect Contact
- D. Indirect Mechanical

ANSWER: A

127. A mother came to a health center with complaints of abdominal cramp pain and watery diarrhea. In history, she explained that they do not have proper latrine facility, hand washing facility, and live in desperate condition. She prepares meal for her child, and served her family. They do not have habits of eating raw and undercooked food. After few days her family members complain the same signs and symptoms of disease.

What is the most likely means of disease transmission in this case?

- A. Food
- B. Finger
- C. Fomites
- D. Flies

ANSWER: B

128. A man who was served food in a local restaurant bites fine sands that were accidentally found in the food. He was seriously injured, especially on his right molar tooth. He called the owner of the restaurant and complained for serving unhygienic food.

What is the most likely type of food safety hazards manifested in this case?

- A. Physical
- B. Biological
- C. Chemical
- D. Allergens

ANSWER: A

129. A six-year-old child came to a health center with complaints of vomiting and diarrhea 24 hours after eating food in the local restaurant. The laboratory investigation result showed that the child has food poisoning caused by spore-forming microbial agents.

What is the most likely cause of the disease in this case?

- A. *Clostridium botulinum*
- B. *Campylobacter jejuni*
- C. *Salmonella*
- D. *Yersinia*

ANSWER: A

130. An environmental health officer who is working in food manufacturing industry wants to establish Conduct Hazard analysis and critical control point (HACCP). He established the team from different units of the industry. The team started to identify the potential hazards associated with food production at all stages from primary production, processing, manufacturing and distribution until the point of consumption to assess the occurrence of the hazards and decided their control measures.

What are the most likely steps of HACCP principle conducted by the team in this case?

- A. Identifying the critical control points
- B. Establishing critical limits
- C. Establishing monitoring procedures
- D. Conducting a hazard analysis

ANSWER: D

131. A health care worker in a health center reported food borne disease outbreak. They identified the source of outbreak was the local restaurant nearby the health center. The woreda health office sent the team of health professionals to confirm the causes; the team recommended the need for quick action until further investigation is conducted.

What is the immediate corrective measure to be taken to prevent further disease transmission?

- A. Conduct an inspection
- B. Take a food sample
- C. Give a health education
- D. Close the restaurant

ANSWER: D

132. A laboratory technician wants to collect up to 125ml of milk sample for laboratory analysis, and he found the milk is in bulk containers. The sample collection should be appropriate, representative, and free from contamination, and bias.

What is the most likely first step the laboratory technician has to do in this case?

- A. Mixing the bulk of milk using a sterile plunger
- B. Taking the sample from the top with a sterile dipper
- C. Taking the sample from the well below with a sterile dipper
- D. Transferring the sample into a sterile stopper or screw capped bottle

ANSWER: A

133. A mother uses the oldest methods to improve food quality; hence, she usually preserves meat and fish using smoking techniques. She exposes the meat and fish to the smoke complimented with adding salt and preserve for months.

What is the most likely physical characteristic of meat and fish that is changed in this case?

- A. Microorganism
- B. Temperature
- C. Moisture
- D. Texture

ANSWER: C

134. A 40-year-old male patient came to a health center with complaints of nausea, vomiting, and bloody diarrhea. He ate eggs and poultry meat in the local restaurant a day before. The food handler working in the restaurant had similar signs and symptoms and she was suspected source of food contamination.

What is the most likely cause of the food borne disease in this case?

- A. Yersinia
- B. Shigella
- C. Salmonella
- D. Staphylococcus

ANSWER: C

135. An environment health officer who is working in the regional health laboratory randomly checked the quality of food ingredients mentioned on labeling of food packages. The officer investigated six different types of food and the laboratory result indicated five of the tested food listed false ingredients that are not present in the food.

What is the most likely safety regulation violated in this case?

- A. Adulteration
- B. Misbranding
- C. Contamination
- D. Wholesomeness

ANSWER: B

136. Large number of people displaced from one woreda due to natural disaster and relocated in the areas covered by indigenous trees and forest land. Even though their settlement is not exactly in the forest, they access the forest within a one-kilometer radius. They started agricultural activities by removing forests, and this condition increased and impacted most parts of the forest. Experts identified that the removal of trees or vegetation was to the extent that it no longer supports its natural flora and fauna. This condition led to extreme consequences and disaster associated with heavy flooding and massive displacement.

What is the most likely cause for disaster in this case?

- A. Deforestation
- B. Man made
- C. Natural
- D. Desertification

ANSWER: A

137. In a semi-urban village, people settled in the lower elevation of the area are frequently affected by heavy rainfall and flooding once a year. Residents usually contract temporary shelter and use the water supply from the neighboring village, and few of them also use the communal latrine. The environmental health officer conducted a survey on the sanitation facilities of residents a month after flooding, and almost 85 percent practice open defecation, which was only 15 percent in the past.

What is the most likely environmental health service affected in this case?

- A. Latrine facility
- B. Drinking water facility
- C. Hand washing facility
- D. Drinking water treatment facility

ANSWER: A

138. Based on a report from a local community leader, an epidemic investigation team arrived to the affected community. The team assessed the situation, the existence of clustered cases and assured that the clusters case is an outbreak.

What is the most appropriate next step taken by the team in this case?

- A. Develop hypothesis
- B. Verify the diagnosis
- C. Construct case definition
- D. Perform descriptive epidemiology

ANSWER: B

139. A report indicated that adults and children in a village were affected by a diarrheal disease. An established health team figured out that the average incubation period for all the cases was similar and the outbreak was caused by a fecal contamination of the drinking water source where those village dwellers were using.

What is the most likely type of epidemics in this case?

- A. Chronic epidemic
- B. Intermittent epidemic
- C. Long lasting epidemic
- D. Point source epidemic

ANSWER: D

140. After detection of a highly contagious disease in patient who was in operation room, the hospital infection prevention officer immediately planned to disinfect the operation room with appropriate disinfectant. However, the room is filled with large size equipment's which cannot be moved or covered during disinfection.

What is the most appropriate disinfectant to be used for this case?

- A. Chlorine solution
- B. Alcohol sublimation
- C. Formaldehyde solution
- D. Hydrogen peroxide fumigation

ANSWER: D

141. A health center's infection prevention and control officer were requested to recommend a disinfectant for hand rubbing in the health center. Based on various criteria, the officer finally chose alcohol as a primary disinfectant.

What are the most likely criteria used to select alcohol in this case?

- A. High Efficacy
- B. Residual effect
- C. Preferred Brand
- D. Sort Contact time

ANSWER: A

142. In an area where open defecation is rampant, dysentery from a specific causative agent occurred as an epidemic. The team of experts, who investigated the epidemic, found out that the disease was propagated due to the nature of transmission and convenient condition for the vector reproduction. It is confirmed that the vector transmitted the causative agent mechanically only by carrying on its body parts without any change happening to the agent.

What is the most likely vector responsible for the transmission of the disease in this case?

- A. Rodents
- B. Mosquito
- C. Bed bugs
- D. House fly

ANSWER: D

143. During a house-to-house visit, an environmental health officer found out that households in a village are infested with cockroaches. After completing the visit, the officer called for community awareness campaign to control the cockroach infestation.

What is the most appropriate sustainable control strategy for this case?

- A. DDT spray
- B. Fumigation
- C. Sanitation
- D. Personal hygiene

ANSWER: C

144. In a country where malaria is endemic for long period of time, the government planned an effective and long-term malaria control mechanism. The national task force designed different tools for the aimed malaria control. What is the most effective tool for this case?

- A. Aerial chemical spray
- B. Educating the public
- C. Eliminate all stagnant water
- D. Mass use of personal protective equipment

ANSWER: B

145. An environmental health officer noticed that a waste dumping site has become a major breeding site for common houseflies. To prevent disease transmitted through flies, the officer wants to spray the area with an insecticide.

What is the most appropriate insecticide formulation needed for this case?

- A. 1.5 percent dieldrin
- B. 2.5 percent chlordane
- C. 3.5 percent endrin
- D. 4.5 percent methoxychlor

ANSWER: B

146. In a highland village, lice infestation is common that it needs the attention of public health professionals. An environmental health officer was assigned to find a solution to the problem, where he identified that applying pesticide was the best immediate approach.

What pesticide formulation could be more convenient to apply on the head of the infested individuals?

- A. Dust
- B. Solution
- C. Wettable powders
- D. Emulsifiable concentrates

ANSWER: A

147. Vector resistance to the effectiveness of pesticides, environmental pollution and public risk are the global challenges that the world is facing in relation to vector control. Various methods are tested to overcome the problems and achieved effective vector control.

What is the most appropriate method to overcome the problems?

- A. Use of biological control methods
- B. Controlling climate change effects
- C. Integrating vector control methods
- D. Minimize environmental contamination

ANSWER: C

148. During an inspection of a dairy product distribution store, an environmental health officer took a 10 mL sample from the milk to check the quality. Then, he added about 1 mL of standard ethylene blue solution and observes the disappearance of the solutions color.

What is the most likely milk quality indicator the officer is checking in this case?

- A. Adulteration
- B. Amount of fat
- C. Bacterial load
- D. Chemical contamination

ANSWER: C

149. During a routine checkup, the inspector detected that the freezing point of the milk served at the hospital is higher than the recommended value for fresh milk.

What is the most likely cause of the milk quality problem in this case?

- A. Less fat content
- B. Presence of chemicals
- C. Adulteration with water
- D. Presence of high bacterial load

ANSWER: C

150. In a town, food illness is becoming a major public health issue. Team of public health experts decided to provide food safety awareness creation campaign to the public, specifically on highly perishable food items.

What food item should be the primary concern of the awareness campaign in this case?

- A. Cooked meats
- B. Foods with high sugar
- C. Dried or pickled foods
- D. Chemically-preserved foods

ANSWER: A

151. A disaster preparedness and mitigation task force in a region was requested to prepare a disaster mitigation strategy for a drought that may happen in some zones of the region. The task force started to organize ideas and gather information to draft the strategy.

What is the most appropriate first step to be considered by the task force in preparing the strategy in this case?

- A. Need assessment
- B. Land use planning
- C. Risk identification
- D. Resource mobilization

ANSWER: C

152. In a village affected by a flood, the community was under panic due to the sudden occurrence of the disaster. Based on the situation, the established disaster management team focus on eliminating the stress and confusion created due to the intensity of the crisis as a primary step.

What is the most likely disaster management style implemented by the team in this case?

- A. Ostrich style
- B. Submissive style
- C. Concessional style
- D. Bargaining style

ANSWER: A

153. Cholera epidemic is becoming a major problem in a town where the community has no basic knowledge on how to control it. The health promotion officer preferred to create an awareness creation campaign through FM radio to mobilize the community for intervention and collaboration.

What is the most likely health education approach applied by the officer?

- A. Information sharing
- B. Group intervention
- C. Social mobilization
- D. Targeted intervention

ANSWER: C

154. In a humanitarian setting, a refugee camp sanitarian wanted to provide hygiene promotion and education. However, humanitarian policy, community readiness and meeting area were not available to provide the intended intervention.

What is the most likely factor lacking in this case?

- A. Predisposing factor
- B. Reinforcing factor
- C. Promoting factor
- D. Enabling factor

ANSWER: D

155. In a humanitarian setting, a refugee camp sanitarian wanted to provide hygiene promotion and education. The sanitarian planned an intervention, but he has no sufficient knowledge of how to implement.

What is the most likely factor that affects the intervention plan of the sanitarian in this case?

- A. Cost
- B. Skill
- C. Culture
- D. Environment

ANSWER: B

156. In a community, significant numbers of diseases are water borne and use water as a vehicle for transmission. For this reason, the environmental health officer wants to check the portability of the water from source until consumption.

What is the most likely water source considered as safe source in this case?

- A. Well water fitted with hand pump
- B. Water obtained from a reservoir
- C. Water transported by a truck
- D. Water transported by a cart

ANSWER: A

157. An environmental health officer wants to bring behavioral change about hygiene and proper utilization of toilets in a selected few household.

What is the most likely suitable specific approach to be applied to bring change in this case?

- A. Film and video
- B. Focus group discussion
- C. One-to-one discussion
- D. Drama and street theatre

ANSWER: C

158. During an assessment of a food complex company, the inspector gathered the non-compliances found and discussed the issue with the manager at the end. The main suggestions given by the inspector were to make the food preparation area non-smoking and to aware food handlers on the proper way of sneezing and coughing.

What is the most likely variable of food service assessment considered in this case?

- A. Health status
- B. Personal Hygiene
- C. Personal behavior
- D. Illness and injuries

ANSWER: C

159. A regional regulatory inspection team planned to conduct an inspection on a previously inspected hotel to assess its compliance with the set standards. The team has decided to evaluate selected service areas (kitchen and toilet) as indicators of the overall compliance of the hotel to the standard.

What is the most appropriate type of inspection to be applied in this case?

- A. Special inspection
- B. Routine inspection
- C. Concise inspection
- D. Follow-up inspection

ANSWER: C

160. Residents of a town are complaining regarding waste management, professional practice and services provided by the health center. The health center decided to deal with specific complaints received about non-compliance with standards of professional practice and provided services.

What is the appropriate type of inspection to be conducted in this case?

- A. Special inspection
- B. Routine inspection
- C. Follow-up inspection
- D. Investigative inspection

ANSWER: D

161. In a food complex company producing different types of food items in a city, the persons who have direct contact in food manufacturing and preparation were found to be victims of food-borne illness. This problem was reported by the inspection team to the city health bureau to take immediate measures accordingly.

What is the most appropriate proclamation to take legal actions in the above scenario?

- A. Proclamation No.759/2012
- B. Proclamation No.1112/2019
- C. Proclamation No.661/2009
- D. Proclamation No.200/2000

ANSWER: B

162. In a town, environmental and social impact assessment (ESIA) firm has been hired by the owner of the project to conduct a full assessment of hydroelectric dam project. As a starting point, the firm has finalized the terms of reference.

What is the most likely next step in conducting the ESIA in this case?

- A. Screening
- B. Reviewing
- C. Decision-making
- D. Impact Analysis

ANSWER: D

163. The environmental health officer working in a district health bureau has recommended that the water and sewerage authorities should do an environmental and social impact assessment (ESIA) study for the rain water harvesting project planned for a city.

What is the most likely type of ESIA required for the project in Ethiopian law?

- A. No ESIA
- B. Full scale ESIA
- C. Conditional ESIA
- D. Preliminary ESIA

ANSWER: D

164. A university is planning to open a new campus in the nearby town. Multidisciplinary team members conducted Environmental and social impact assessment (ESIA) and submitted the report to the zonal Environmental Protection Authority (EPA). The EPA reviewed the report and came to the conclusion that the project may have impact on the reproductive capacity of the local environment in a way it can no longer gain it back.

What is the most appropriate type of impact the project can have in this case?

- A. Reversible impact
- B. Significant impact
- C. Cumulative impact
- D. Irreversible impact

ANSWER: D

165. An international company is planning to construct an irrigation-based agricultural project on 40 hectares of land. Before receiving a trade license, the company was requested to get approval from the regional Environmental Protection Authority (EPA) on the bases of environmental and social impacts.

What is the most likely decision that can be made by the EPA?

- A. Approval after full scale ESIA
- B. Direct approval without ESIA
- C. No decision can be made at this level
- D. Temporary approval with preliminary ESIA

ANSWER: B

166. A new project, submitted to the national Environmental Protection Authority (EIA) for approval, has undergone a review process before being sent to the Environmental and social impact assessment (ESIA) study. However, the team of experts gave clearance for the project implementation without an ESIA. What is the most likely evidence for the team's decision in the above case?

- A. The project may have probable impacts
- B. The project is very useful to the community
- C. The project is in line with the government policies
- D. The project is unlikely to cause significant environmental impacts

ANSWER: D

167. The head of the municipal hygiene and sanitation unit wants to prepare a five year strategic plan to have well designed waste management system. To do so, the head assessed population size of the town, magnitude of the problem, availability of donors and feasibility of the plan.

What is the most likely planning step conducted by the head in this case?

- A. Setting objectives
- B. Situational analysis
- C. Strategic analysis
- D. problem prioritization

ANSWER: B

168. To prepare a strategic plan, the planning team of a hospital conducted situational analysis and identified many problems. To include all identified problems in the plan, the team has resource constraints; some problems are technically difficult, and some other problems are the priority of the health policy.

What are the most appropriate prioritization criteria to be applied in this case?

- A. Severity of problems and community concern
- B. Magnitude of problem and sustainability
- C. Feasibility and government concern
- D. Magnitude of problem and severity of problems

ANSWER: C

169. The zonal health department assigned an environmental health officer as a supervisor for health extension workers to coordinate their activities and support them while providing quality health services for the population in the catchment area.

What are the most important managerial skills for the environmental health officer in this case?

- A. Human relations and technical skills
- B. Conceptual skill and human relations
- C. Conceptual skill and technical skills
- D. Strategic thinking skill and technical skills

ANSWER: A

170. The most assertive and cooperative, department head solve interpersonal conflicts through the involvement of others and attending fully to their concerns without sacrificing or suppressing his own concerns.

What is the most likely conflict resolution strategy applied by the manager in this case?

- A. Collaboration
- B. Accommodation
- C. Competition
- D. Avoidance

ANSWER: A

171. The leader at the regional health bureau uses a unique leadership style. More importantly the leader believes and applies the principles of changing the attitudes, values, and behaviors of staff in ways that align with organizational goals which will bring better organizational success.

What is the most likely leadership style applied by the leader?

- A. Transformational
- B. Transactional
- C. Charismatic
- D. Visionary

ANSWER: A

172. The head of zonal health department made a decision about the next year's priority area. During the decision-making process, the head used the results of the previous implementation performance and determined the desired individual and/or organizational results.

What is the most likely decision-making method used by the head?

- A. Ends decision
- B. Means decision
- C. Administrative
- D. political decision

ANSWER: A

173. The medical director asked all the units in the hospital to bring the materiel requirements for the next year's procurement process. All units listed the material needed and gave it to the medical director to compile and prepare the requisition and approve it for purchasing.

What is the most likely material management component performed by the director in this case?

- A. Ordering
- B. Issuing
- C. Purchasing
- D. inventory

ANSWER: A

174. The human resource manager in a referral hospital aspires to a good human resource management system for better performance of the organization. The manager considered all procedures for acquiring and retaining mechanisms for human resources.

What is the most appropriate human resource management principle to be considered first in this case?

- A. Job analysis and job description
- B. Human resource planning
- C. Advertising the vacancy
- D. Selecting the best person

ANSWER: A

175. The government of Ethiopia considered different financial sources to ensure equity through accessing health services for the poor, street children, homeless citizens and displaced persons.

What is the most likely financial source that helps to cover the costs of these individuals?

- A. Health insurance system
- B. Fee Waiver system
- C. Exemption System
- D. User fee setting system

ANSWER: B

176. A medical doctor was assigned as head of specialized hospital in Addis Ababa city. Within the three-tier health system, arrangement the capacity of health facilities is specified for each level of hospitals and also different for urban and rural health centers.

What is the most likely capacity of specialized hospital in three tier health system?

- A. 3.5-5.0 million people
- B. 1-1.5 million people
- C. 60,000-100,000 people
- D. 15,000-25,000 people

ANSWER: A

177. A project head who works on hygiene and sanitation needs to know the contributions of the project on awareness, practice and attitude change of the community towards open field defecation. What is the most appropriate indicator of program evaluation to be considered in this case?
- A. Output indicator
 - B. Outcome indicator
 - C. Process indicator
 - D. Impact indicator
- ANSWER: B
178. An evaluation team planned to conduct evaluation on TB- HIV project so as to know the compliance of interventions to the plan. And it measured the resources used, the activities done; the immediate results achieved and attempted to justify the reasons for their success. What is the most likely program monitoring and evaluation activity to be done in this case?
- A. Input output monitoring
 - B. Process evaluation
 - C. Outcome evaluation
 - D. Impact evaluation
- ANSWER: B
179. During review of a new waste management strategy for a city, the management board suggested that instead of focusing on disposal, the new strategy must focus on reducing, reusing, recycling and recovery so that the waste management will not only benefit the current generation but also maintains the resources for the future generation. What is the most likely environmental ethics principle addressed in this case?
- A. Realism
 - B. Global equity
 - C. Healthfulness
 - D. Sustainability
- ANSWER: D
180. An executive director of a teaching hospital evaluated the health workers performance based on professional standard evaluation criteria set for the purpose. During the assessment, two staff members from similar discipline were excellent according to most of the evaluation criteria. However, they have never got training nor upgraded their education level since employment. What are the most likely criteria the two professionals failed to attain in this case?
- A. Professional independence
 - B. Promotion of the Profession
 - C. Continuous professional development
 - D. Clause on ethics in contracts of employment
- ANSWER: C
181. A team of researchers from various disciplines decided to initiate research that involves animals as test subjects. However, after they prepare the protocol, they were asked by the institutional review board to justify the ethical correctness of using animals for this research. What is the most appropriate justification for the above scenario?
- A. Animals are used to replace the use of human in the research
 - B. No other option is available which can give the same result
 - C. Using laboratory animals in research is the best ethical approach
 - D. Animals used in the research are test animals which has no other economic value
- ANSWER: B

182. A district health manager uses criteria to evaluate the performance of health workers in his catchment. Accordingly, an environmental health practitioner working at a health center got high score on conduct related to maintaining high professional standard through scientific knowledge and technical ability.

What is the most likely professional characteristic described in this case?

- A. Competency
- B. Accountability
- C. Confidentiality
- D. Responsibility

ANSWER: A

183. An environmental health officer and his team were conducting a water source and distribution inspection as a response to the ongoing cholera epidemic. During laboratory analysis, the environmental health officer found out that one of the water sources is contaminated from fecal sources. Consequently, he decided on the closure of the water source by himself. However, his team members and the community using the water sources did not know why this decision was made.

What is the most likely principle of ethical decision making overlooked in this case?

- A. Justice
- B. Empathy
- C. Transparency
- D. Accountability

ANSWER: C

184. A health worker is preparing for consecutive health information dissemination to the health center clients. He mastered the subject matter and prepared the core message beforehand. He finally starts to encode the message to transfer into a format that can be shared to the clients.

What is the most appropriate knowledge required by the health worker at this stage?

- A. Knowing the audience
- B. What is the result expected?
- C. Why the communication is necessary
- D. Acquiring additional related knowledge

ANSWER: A

185. In one district water borne diseases have been highly prevalent for consecutive years. Consequently, the district health and water supply offices organized an information dissemination campaign for the community so that they will understand the strong and weak sides of the available water sources in the area before they decide which source to use.

What is the most likely communication approach used in this case?

- A. Informing
- B. Education
- C. Persuading
- D. Entertaining

ANSWER: B

186. In one region, a confirmed meningitis disease epidemic had occurred, and the government is curious that transmission may occur faster. While doing other preparation works, the ministry of health in the country used radio and television to aware people about the epidemic, its symptoms, and prevention methods.

What is the most likely reason for choosing this communication method in this case?

- A. It can deliver the message clearly
- B. It can reach large population faster
- C. It is cheap to prepare and disseminate
- D. It can deliver a message understood by all group

ANSWER: B

187. During a quarterly review of a health center's performance, the management raised an idea to modify the focus of health information at a health center from 'family planning' to 'communicable disease control'. Each group from the available specialties forwarded their opinion before concluding the focus of discussion to 'maternal and child health for a better society'.

What is the most likely component of communication discussed in this case?

- A. Effect
- B. Source
- C. Channel
- D. Message

ANSWER: D

188. A district health office distributed insecticide-treated bed net to the catchment population for the last five years; however, the occurrences of malaria within this population were not reduced. Accordingly, the district health office administrator invited an environmental health officer to conduct research to exploring why and how the cultural, behavioral, and societal believe affects the insecticide treated bed net utilization.

What is the most appropriate type of research to be used in this case?

- A. Basic research
- B. Fundamental research
- C. Quantitative Research
- D. Qualitative research

ANSWER: D

189. The prevalence of hypertension at national level was dramatically increasing from time to time, and a ministry of health wants to conduct research to assess the relationship of hypertension rates and average per capita salt consumption at different district to design an effective intervention.

What is the most appropriate study design to be used for this case?

- A. Cohort study
- B. Ecological study
- C. Case-control study
- D. Cross-sectional study

ANSWER: B

190. A researcher wants to conduct research to investigate the cause of trachoma. The researcher identified individuals diagnosed with trachoma in a hospital and trachoma free individuals selected from among members of the community served by this hospital at the same time. Finally, the researcher ascertains the cause of trachoma retrospectively.

What is the most likely study design used for this case?

- A. Cross-sectional study
- B. Case-control study
- C. Prospective cohort study
- D. Retrospective cohort study

ANSWER: B

191. An environmental health expert studied the magnitude of malaria in a district. Since malaria prevalence is different with districts an environmental health expert categorizes the district as lowland, midland and high land climatic agro-ecologic condition and the total sample size was proportionally distribute to each district.

What is the most likely sampling technique used by environmental health expert for this case?

- A. Cluster sampling
- B. Stratified sampling
- C. Simple random sampling
- D. Systematic random sampling

ANSWER: B

192. A researcher studied households' waste management practice in a city characterized with scattered households with number of Kebele's. The researcher included around 30 percent of the Kebele's to save money and time by considering that the information got from those kebeles is homogeneous.

What is the most likely type of sampling technique used by the researcher in this case?

- A. Cluster sampling
- B. Stratified sampling
- C. Simple random sampling
- D. Systematic random sampling

ANSWER: A

193. A survey is planned to determine the proportion of malaria in a community. To determine the appropriate sample size; the research team decided to estimate the malaria by taking an assumption 95 percent confidence level and 0.05 margin of error. However, the estimate of proportion malaria is not available from previous studies; a pilot sample cannot be drawn due to various reasons.

What is the recommended proportion to be used by the researcher to estimate the sample size?

- A. 0.20
- B. 0.30
- C. 0.40
- D. 0.50

ANSWER: D

194. An environmental health officer wants to assess the food hygiene practice of food and beverages in a town. The officer identified 30 percent of the restaurants in the town to assess the food handler practice through inspection. What is the most appropriate data collection tool for this case?

- A. Photo
- B. Video
- C. Checklist
- D. Questionnaire

ANSWER: C

195. A medical research team wants to assess usefulness of fever symptom to the diagnosis of malaria. The research team randomly sample patients with confirmed malaria disease and majority reported having a symptom of fever, and from the independent sampled individuals without malaria; very few of them report that they had the symptom of fever.

What is the most likely used data source to the research this case?

- A. Census
- B. Survey
- C. Health recoded
- D. Vital registration

ANSWER: B

196. A researcher designed research to assess gastric disease and blood type. The researcher identified subjects with gastric disease and non-gastric individuals to investigate their blood type.

What is the most likely measurement scale used by the researcher in this case?

- A. Ratio
- B. Nominal
- C. Ordinal
- D. Interval

ANSWER: B

197. A researcher designed an experiment to assess the effects of prolonged inhalation of cadmium oxide on hemoglobin level on experiment animal. Fifteen laboratory animals served as experimental subjects, while 10 similar animals served as controls to assess hemoglobin level following the experiment.

What is the most likely outcome variable in this research?

- A. Hemoglobin level
- B. Laboratory animals
- C. Prolonged inhalation
- D. Effect of cadmium oxide

ANSWER: A

198. In a district hospital; 13 HIV-positive patients who were treated with active Antiretroviral therapy for at least 6 months. The CD4 T cell counts at baseline for the 13 subjects are listed as; 30, 58, 200, 103, 313, 142, 205, 245, 427, 395, 345, 170, and 205.

What is the mode of CD4 T cell counts HIV-positive patients in this case?

- A. 245
- B. 187
- C. 205
- D. 427

ANSWER: C

199. A researcher wants to know age of mothers at first birth in community. The researcher includes 30 mothers in the study and found the grouped age distribution of live births as; 10 mothers in a group of 15 to 19, 5 mothers in a group of 20 to 24, 10 mothers in a group of 25 to 29, and 5 mothers in group of 30 to 34-years-old.

What is the mean age of the mother in this case?

- A. 28.0
- B. 23.7
- C. 30.0
- D. 25.5

ANSWER: B

200. A researcher studied the relationship between body weight (kg) and resting metabolic rate (kcal/24 hours) of adult individuals in a community.

What is the most likely recommended method of data presentation for this data?

- A. Bar graph
- B. Line graph
- C. Histogram
- D. Scatter plot

ANSWER: D

